through time, lassos a two-dimensional sphere. Figure 13.1 should be viewed as a snapshot taken at one instant in time.) Similarly, in Figure 13.1 we see that a two-dimensional membrane—a two-brane—can wrap around and completely cover a two-dimensional sphere, much as a piece of plastic wrap can be tightly wrapped around the surface of an orange. Although it's harder to visualize, Strominger followed the pattern and realized that the newly discovered three-dimensional ingredients in string theory—the three-branes—can wrap around and completely cover a three-dimensional sphere. Having seen clear to this insight, Strominger then showed, with a simple and standard physics calculation, that the wrapped three-brane provides a tailor-made shield that exactly cancels all of the potentially cataclysmic effects that string theorists had previously feared would occur if a three-dimensional sphere were to collapse.

This was a wonderful and important insight. But its full power was not revealed until a short time later.

Tearing the Fabric of Space—with Conviction

One of the most exciting things about physics is how the state of knowledge can change literally overnight. The morning after Strominger posted his paper on the electronic Internet archive, I read it in my office at Cornell after having retrieved it from the World Wide Web. In one stroke, Strominger had made use of the exciting new insights of string theory to resolve one of the thorniest issues surrounding the curling up of extra dimensions into a Calabi-Yau space. But as I pondered his paper, it struck me that he might have worked out only half of the story.

In the earlier space-tearing flop-transition work described in Chapter 11, we had studied a two-part process in which a two-dimensional sphere pinches down to a point, causing the fabric of space to tear, and then the two-dimensional sphere reinflates in a new way, thereby repairing the tear. In Strominger's paper, he had studied what happens when a three-dimensional sphere pinches down to a point, and had shown that the newfound extended objects in string theory ensure that physics continues to be perfectly well behaved. But that's where his paper stopped. Might it be that there was another half to the story, involving, once again, the tearing of space and its subsequent repair through the reinflation of spheres?